DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics, and Computer Science

Dynamic Programming TA Review: Pretty Printing

CSE2310 Algorithm Design

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Deadline: 20 December 23:59[1]

Report

For this exercise, we expect you to write a report in **your own words**. Your report should meet the following constraints:

- It should be a **PDF** file with at most two sides of an A4 (excluding an optional front page or space used by diagrams). The use of LaTeX is recommended.
- It should be written in clear, readable, and correct Dutch or English.
- It should contain **ONLY** your student number (**NOT** your name), the exercise number, and
- for each question
  - a clear and brief explanation (intuition) about why your answer/(pseudo)code is correct,
  - a reference to the material (book/slides), why this applies here, what you changed, and a reference to the relevant part of your code (all only if applicable),
  - a reference to the students with whom you conferred to obtain this solution (if this is the case), and
  - an indication where you have doubts or are stuck.

General recommendations and hints

- Read the complete assignment carefully before you start solving the subproblems.
- If you get stuck: don't panic, but read the relevant sections in the book first. You can also ask the assistants to help you.

[1] This is the advised time for submission. Submission via BrightSpace is possible for a limited time period after this due date.

In a word processor, the *pretty-printing* function takes a text with ragged right margin, like this:

```
Marley was dead: to begin with.
There is no doubt whatever about that.
The register of his burial was signed by the clergyman,
the clerk, the undertaker, and the chief mourner.
Scrooge signed it: and Scrooge's name was good upon 'Change,
for anything he chose to put his hand to.
Old Marley was as dead as a door-nail.
```

and turns it into text with a right margin which is as "even" as possible, like this:

```
Marley was dead: to begin with. There is no doubt whatever about
that. The register of his burial was signed by the clergyman,
the clerk, the undertaker, and the chief mourner. Scrooge signed
it: and Scrooge's name was good upon 'Change, for anything he
chose to put his hand to. Old Marley was as dead as a door-nail.
```

To make the problem more exact, we introduce some notation. A text consists of a sequence of n words, $W = \{w_1, w_2, \ldots, w_n\}$, where w_i consists of c_i characters. The maximum line length is L characters, including spaces. Any punctuation is part of the words, and all characters have equal width (i.e., a fixed-width font is used). Words may not be hyphenated.

A formatting of W is a partition of the words in W into lines. After each word in a line, except the last, a single space is added. Therefore, if $w_i, w_{i+1}, \ldots, w_j$ are assigned to a line, it must hold that

$$\left[\sum_{k=i}^{j-1}(c_k + 1)\right] + c_j \leq L.$$

A line for which this inequality holds is called a *valid line*.

The difference between the left-hand side and the right-hand side of this inequality is called the *slack* of the line. The goal of the pretty-printing function is to find a partition of a sequence of words W into valid lines, such that the sum of the *squares* of the slacks of all lines, including the last, is minimized.

1. As a starting point in solving the problem above, let $S_{i,j}$ represent the slack of a line containing $\{w_i, \ldots, w_j\}$, for $i \leq j$; let $S_{i,j} = \infty$ if the length of the line exceeds the total length L. Give pseudocode to compute $S_{i,j}$ for all $1 \leq i \leq j \leq n$ given the word lengths $c_1, c_2, \ldots, c_n$.

2. Give a *recursive formula* that returns the sum of squared slacks for the optimal partition of a sequence of words W into lines of up to L characters (including spaces). Use $S_{i,j}$ as defined in the question above.

3. Give pseudocode for an efficient iterative algorithm using the dynamic programming approach and the function from the previous question to determine the sum of squared slacks of the optimal solution for the set of words W, using line length L. You may assume $S_{i,j}$ is already calculated.

4. Give the tightest upper bound on the run time of your algorithms (i.e., the combined run time of both your algorithm in question 1 and question 3), and briefly explain your answer.

5. Give pseudocode for an algorithm that uses the result of the previous algorithms to efficiently pretty-print the sequence of words W. You may assume a function print-line(P) exists, which outputs the sequence of words P with spaces in between, on a single line.